

CAPSTONE PROJECT MODULE DESIGN

Clinic Management System

Module Design, Workflows, Database Structures, Reports, and Access Controls

Document Type	System Module Design Document
Scope	Ten integrated clinic management modules
Prepared For	College Capstone Project Documentation
Version	1.0
Date	August 2026

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1. System Overview

The Clinic Management System centralizes student and employee health services, clinic operations, medicine management, appointments, emergency cases, health programs, certificates, reporting, and confidentiality controls. The design is intended for a school setting and supports accountable, role-based processing of sensitive health information.

System Design Principles

Principle	Design Application
Integrated records	Medical, visit, medicine, appointment, and certificate data are connected through patient and user identities.
Workflow control	Each operational module uses a defined sequence to minimize incomplete, duplicate, or unauthorized processing.
Traceability	Key actions are retained through transaction records and audit logs.
Confidentiality	RBAC and privacy safeguards limit access to sensitive health information.

2. Module Design Summary

Module	Purpose	Primary Users
1. Student Medical Records Management	Stores and manages the complete health records of every student.	Clinic Staff; School Nurse; Doctor; Administrator
2. Clinic Visit and Consultation Logging	Records each student visit from check-in through check-out and preserves visit history.	Clinic Staff; School Nurse; Doctor; Administrator
3. Medicine Inventory and Dispensing	Tracks clinic medicines, stock movements, expirations, and dispensing transactions.	Administrator; Clinic Staff; School Nurse; Doctor
4. Appointment Scheduling System	Allows students to reserve and manage clinic appointments.	Student; Clinic Staff; Doctor; Administrator
5. Incident and Emergency Case Management	Handles emergency situations occurring within the school and records care, notifications, and referrals.	Clinic Staff; School Nurse; Doctor; Administrator
6. Faculty and Staff Health Services	Provides clinic services and health documentation for employees.	Faculty and Staff; Clinic Staff; School Nurse; Doctor; Administrator
7. School Health Program Monitoring	Monitors the planning, participation, assessment, and completion of school health programs.	Clinic Staff; School Nurse; Doctor; Administrator; Students
8. Health Clearance and Certification	Generates, approves, verifies, and stores health-related certificates and clearances.	Student; Clinic Staff; School Nurse; Doctor; Administrator
9. Reporting and Compliance	Provides operational analytics, printable reports, and compliance-ready summaries.	Administrator; Clinic Staff; School Nurse; Doctor
10. User Access and Confidentiality Control	Protects confidential medical information through role-based access control (RBAC), auditability, and data-security controls.	Administrator; Clinic Staff; School Nurse; Doctor; Student; Faculty and Staff

3. Detailed Module Specifications

1. Student Medical Records Management

Design Element	Specification
Purpose	Stores and manages the complete health records of every student.
Users	Clinic Staff; School Nurse; Doctor; Administrator
Key Features	Student registration; medical history; allergies; immunization records; chronic diseases; height and weight monitoring; vision and hearing test; blood type; family medical history; PDF/image attachments.
Reports	Student Medical Profile; Students with Allergies; Students with Asthma; Immunization Status Report; Medical History Report.
Security / Access	Authorized clinical users can view and update records according to role permissions; attachments and health history are confidential.

Process / Workflow

Step	Workflow Activity
1	Student enrolls
2	Student information is imported or encoded
3	Clinic creates medical record
4	Medical information is entered
5	Medical record becomes active
6	During each visit, profile, diagnosis, treatment, and history are updated

Database Tables and Fields

Table	Fields
Students	StudentID; StudentNo; FullName; Course; YearLevel; Section; ContactNo; Guardian
MedicalRecords	RecordID; StudentID; BloodType; Allergies; MedicalConditions; Height; Weight; Vision; Hearing; ImmunizationStatus; Remarks
MedicalHistory	HistoryID; StudentID; Diagnosis; Treatment; Doctor; VisitDate

2. Clinic Visit and Consultation Logging

Design Element	Specification
Purpose	Records each student visit from check-in through check-out and preserves visit history.
Users	Clinic Staff; School Nurse; Doctor; Administrator
Key Features	Check-in; queue number; reason for visit; vital signs; consultation; diagnosis; medicine dispensing; check-out; visit-history search.
Reports	Daily Patients; Weekly Patients; Monthly Visits; Most Common Illnesses; Student Visit History.
Security / Access	Only authorized staff may manage a visit. Vital signs precede consultation; completed visits are retained and actions are audit logged.

Process / Workflow

Step	Workflow Activity
1	Student arrives
2	Staff searches student and checks in
3	System generates queue number
4	Nurse records vital signs
5	Doctor conducts consultation
6	Diagnosis and treatment are encoded
7	Medicine is released when applicable
8	Visit is completed and history is saved

Database Tables and Fields

Table	Fields
Visits	VisitID; StudentID; VisitDate; Reason; QueueNo; Status
VitalSigns	Temperature; BloodPressure; Pulse; Respiration; Height; Weight
Consultation	Diagnosis; Treatment; Doctor; Notes

3. Medicine Inventory and Dispensing

Design Element	Specification
Purpose	Tracks clinic medicines, stock movements, expirations, and dispensing transactions.
Users	Administrator; Clinic Staff; School Nurse; Doctor
Key Features	Medicine registration; supplier and batch details; expiration tracking; stock monitoring; low-stock alerts; issuance; disposal.
Reports	Medicine Inventory; Expired Medicines; Near Expiration; Most Dispensed Medicine; Low Stock; Medicine Usage.
Security / Access	Dispensing is restricted to authorized users. Available stock is checked before release; inventory adjustments are recorded.

Process / Workflow

Step	Workflow Activity
1	Medicine arrives
2	Administrator records stock
3	Medicine is stored as available stock
4	Doctor prescribes after consultation
5	Staff dispenses medicine
6	Quantity is deducted automatically
7	Transaction is recorded
8	Low stock triggers purchase request

Database Tables and Fields

Table	Fields
Medicines	MedicineID; Name; Category; Unit; Stock; ReorderLevel; ExpirationDate; Supplier
Dispensing	DispenseID; StudentID; MedicineID; Quantity; DateReleased; ReleasedBy

4. Appointment Scheduling System

Design Element	Specification
Purpose	Allows students to reserve and manage clinic appointments.
Users	Student; Clinic Staff; Doctor; Administrator
Key Features	Appointment booking; doctor availability; time slots; approval; reschedule; cancellation; email/SMS notification.
Reports	Today's Appointments; Missed Appointments; Completed Appointments; Cancelled Appointments.
Security / Access	Students manage only their own requests. Clinic staff and doctors manage availability and approval according to assigned permissions.

Process / Workflow

Step	Workflow Activity
1	Student logs in
2	Select date
3	Select available doctor
4	Choose time slot
5	Submit request
6	Clinic approves
7	Appointment is confirmed
8	Reminder is sent
9	Consultation is completed

Database Tables and Fields

Table	Fields
Appointments	AppointmentID; StudentID; DoctorID; Date; Time; Status; Reason
DoctorSchedule	DoctorID; Date; AvailableTime; Status

5. Incident and Emergency Case Management

Design Element	Specification
Purpose	Handles emergency situations occurring within the school and records care, notifications, and referrals.
Users	Clinic Staff; School Nurse; Doctor; Administrator
Key Features	Incident reporting; student identification; emergency treatment; guardian notification; hospital referral; case closure; incident report.
Reports	Incident Summary; Emergency Cases; Referral Report; Accident Statistics.
Security / Access	Emergency cases are visible only to authorized clinical and administrative roles; all updates remain traceable.

Process / Workflow

Step	Workflow Activity
1	Emergency is reported
2	Staff creates incident
3	Student is identified
4	Emergency treatment is administered
5	Guardian is notified
6	Hospital referral is made if needed
7	Incident is closed
8	Incident report is generated

Database Tables and Fields

Table	Fields
Incidents	IncidentID; StudentID; IncidentType; Location; Severity; ReportedBy; Status
EmergencyTreatment	Treatment; Medicine; Nurse; Date; Remarks

6. Faculty and Staff Health Services

Design Element	Specification
Purpose	Provides clinic services and health documentation for employees.
Users	Faculty and Staff; Clinic Staff; School Nurse; Doctor; Administrator
Key Features	Employee medical record; annual physical exam; consultation; medicine dispensing; medical certificate; leave recommendation.
Reports	Employee Visits; Medical Certificates; Employee Health Summary.
Security / Access	Employees can access only their own records. Clinical data is controlled by role and certificate issuance requires authorization.

Process / Workflow

Step	Workflow Activity
1	Employee visits clinic
2	Staff searches employee
3	Medical consultation is conducted
4	Diagnosis is entered
5	Medicine is dispensed if needed
6	Medical certificate is issued when appropriate
7	Record is saved

Database Tables and Fields

Table	Fields
Employees	Employee identity and employment information
EmployeeMedicalRecord	Employee health profile and medical history
EmployeeVisit	Visit details; diagnosis; treatment; provider; date
EmployeeMedicine	Medicine release details; quantity; release date; released by

7. School Health Program Monitoring

Design Element	Specification
Purpose	Monitors the planning, participation, assessment, and completion of school health programs.
Users	Clinic Staff; School Nurse; Doctor; Administrator; Students
Key Features	Vaccination; deworming; dental check-up; eye check-up; mental health; nutrition program; health seminars.
Reports	Program Attendance; Vaccination Coverage; Health Program Completion; Student Participation.
Security / Access	Program coordinators manage schedules and attendance; participant health assessments remain available only to authorized users.

Process / Workflow

Step	Workflow Activity
1	Create program
2	Schedule program
3	Assign participants
4	Record attendance
5	Conduct health assessment
6	Generate summary report

Database Tables and Fields

Table	Fields
HealthPrograms	ProgramID; ProgramName; Type; Schedule; Coordinator; Status
Participants	ParticipantID; ProgramID; StudentID; Eligibility; Status
Attendance	AttendanceID; ProgramID; ParticipantID; AttendanceDate; Status
Assessment	AssessmentID; ProgramID; ParticipantID; Result; Remarks; AssessedBy

8. Health Clearance and Certification

Design Element	Specification
Purpose	Generates, approves, verifies, and stores health-related certificates and clearances.
Users	Student; Clinic Staff; School Nurse; Doctor; Administrator
Key Features	Medical clearance; fit-to-study; sports clearance; OJT clearance; return-to-school clearance; medical certificate; QR verification.
Reports	Issued Certificates; Pending Certificates; Certificate History.
Security / Access	Certificates require authorized approval. QR codes support verification, while history preserves an auditable record of issuance.

Process / Workflow

Step	Workflow Activity
1	Student requests clearance
2	Clinic reviews records
3	Medical examination is completed
4	Doctor approves
5	Certificate is generated
6	QR code is generated
7	Certificate is printed
8	Certificate is saved to history

Database Tables and Fields

Table	Fields
Certificates	CertificateID; StudentID; Type; IssuedDate; Doctor; Status; QRCode

9. Reporting and Compliance

Design Element	Specification
Purpose	Provides operational analytics, printable reports, and compliance-ready summaries.
Users	Administrator; Clinic Staff; School Nurse; Doctor
Key Features	Dashboard metrics: today's patients, available and expired medicines, low stock, appointments, emergencies, health programs. Analytics: common disease, top medicines, clinic visits, age/course/gender distribution, medicine consumption.
Reports	Daily Report; Weekly Report; Monthly Report; Annual Report; Medicine Report; Inventory Report; Consultation Report; Student Health Report.
Security / Access	Reports are limited by role. Sensitive health details are included only for authorized users and report access should be audit logged.

Process / Workflow

Step	Workflow Activity
1	Operational data is captured by core modules
2	Dashboard aggregates current metrics
3	User selects reporting period and filters
4	System generates analytics and printable report
5	Authorized user reviews or exports report

Database Tables and Fields

Table	Fields
Reporting Views / Aggregates	Derived from visits, medicines, appointments, incidents, certificates, health programs, and audit logs
ReportParameters	ReportType; DateRange; Filters; RequestedBy; GeneratedAt

10. User Access and Confidentiality Control

Design Element	Specification
Purpose	Protects confidential medical information through role-based access control (RBAC), auditability, and data-security controls.
Users	Administrator; Clinic Staff; School Nurse; Doctor; Student; Faculty and Staff
Key Features	Secure login; RBAC; encrypted passwords; session timeout; audit trail; optional two-factor authentication; data encryption; QR verification; backup and restore; activity logs; consent tracking; account lockout.
Reports	User Activity Log; Login Attempts; Role and Permission Matrix; Audit Trail; Backup Status; Privacy Consent Status.
Security / Access	Administrator manages users, roles, backups, and reports. Clinic Staff registers patients and visits; Nurse updates records; Doctor diagnoses and approves certificates; Students and employees access only their own information.

Process / Workflow

Step	Workflow Activity
1	User logs in
2	System authenticates credentials
3	Role is identified
4	Permissions are loaded
5	Authorized modules become accessible
6	All activities are recorded in the audit trail

Database Tables and Fields

Table	Fields
Users	UserID; Username; PasswordHash; RoleID; Status; LastLogin; FailedLoginCount
Roles	RoleID; RoleName; Description
Permissions	PermissionID; Module; Action; Description
RolePermissions	RoleID; PermissionID
AuditLogs	AuditID; UserID; Action; Module; RecordID; Timestamp; IPAddress
PrivacyConsents	ConsentID; SubjectID; ConsentType; GrantedAt; Status

4. Overall Integrated Workflow

The following sequence shows how the principal modules interact during a typical service cycle. Appointments are optional; emergency cases may enter directly through incident management.

Step	Integrated Workflow
1	Student or Employee registers / logs in
2	Medical record is created or retrieved
3	Appointment is booked when needed
4	Clinic visit or emergency case is recorded
5	Vital signs and consultation are completed
6	Diagnosis and treatment are saved
7	Medicine is dispensed when applicable
8	Certificate or clearance is issued when needed
9	Reports and analytics are generated
10	Audit trail and record archive preserve accountability

5. Role and Access Summary

Role	Authorized Responsibilities
Administrator	Manage users, roles, system configuration, backups, reports, and audit logs.
Clinic Staff	Register patients, record visits, dispense medicine, and schedule appointments.
School Nurse	Conduct nursing consultation, record vital signs, and update medical records; issue clearances when authorized.
Doctor	Diagnose patients, prescribe medicines, review history, and approve medical certificates.
Student	View personal medical profile, book appointments, view appointment history, and download approved certificates.
Faculty and Staff	View personal health records, request appointments, and download medical certificates.